



OPEN

DATA DESCRIPTOR

# Multiclass Dataset for Intelligent Detection of Wind Turbine Blade Defects Using Drone Imagery

Lipeng Ji<sup>1</sup>✉, Junjie Cheng<sup>1</sup> & Shilong Wu<sup>1,2</sup>

Achieving intelligent and automated detection of defects in wind turbine blades has become a critical task for contemporary wind farm inspection operations. However, existing datasets for blade defect detection exhibit notable shortcomings, including insufficient defect attributes and limited scale, which hinder the advancement of related detection algorithms. This paper presents a standardized multiclass dataset of visible images of wind turbine blade defects for visual inspection, comprising six categories and 1,065 real blade images captured by unmanned aerial vehicles (UAVs). To provide a comprehensive characterization of this dataset, we conducted a feature space analysis using t-SNE to identify unique attributes of the defective targets. The dataset addresses the lack of diverse defect types and high-resolution samples in existing resources, providing a benchmark for the development of visual inspection algorithms.

## Background & Summary

Wind energy is one of the most essential sustainable and renewable energy sources, and ensuring the efficient and economical operation of wind power generation is a critical task in the current needs of the wind energy industry<sup>1</sup>. It requires non-contact and non-destructive wind turbine (WT) blade detection technology with high accuracy and long-range capabilities. While machine vision-based technologies have emerged to solve this problem<sup>2,3</sup>, an in-depth analysis of these methodologies dramatically depends on the quality of benchmark images<sup>4</sup>. The quality of WT blade images, which provide detailed structural information for blade surfaces, directly affects the detection performance of algorithms, especially in deep learning<sup>5,6</sup>. Generally, defective areas contain specific local features such as color, texture, and edge, which differ from the background. However, due to complex geometrical structures and spatial patterns in engineering application scenarios, dataset images must contain higher-level semantic content to be effectively understood.

As the appearance and physical characteristics of WT blades are highly variable, detection of surface defects is error-prone. There are two main challenges in automated detection: constructing a high-quality blade defect dataset and selecting appropriate detection methods. Previous studies are surveyed in Table 1 to explain vision-based detection technologies more precisely. Early studies employed traditional machine vision methods<sup>7,8</sup>, where artificial feature operators were designed to extract identifiable features based on texture, color, and geometry, followed by classification<sup>9–11</sup>. Familiar operators include the Local Binary Pattern (LBP)<sup>12</sup>, which has the advantage of grayscale and rotation invariance, and the Histogram of Oriented Gradients (HOG)<sup>13</sup>, used to characterize local gradient direction and intensity. Literature<sup>7,8</sup> utilized HOG and Haar-like features to characterize defects, subsequently employing Support Vector Machine (SVM)<sup>14</sup> and extended cascading classifiers for identification. However, limited by the feature extraction capabilities of traditional methods, these relevant datasets generally do not possess many defect types.

More recently, a series of deep learning algorithms have been applied to this task<sup>15–19</sup>. For multiple defect types, literature<sup>15,18</sup> utilized two-stage detection models like Faster-RCNN<sup>20</sup> and Mask R-CNN<sup>21</sup> to complete localization and identification. These models perceive critical information by generating regional candidate frames and performing convolutional operations. Conversely, literature<sup>16,19</sup> used one-stage models<sup>22–24</sup> as a framework to address issues like small targets and detection speed. By constructing YSODA and YOLOv3 algorithms, feature pyramids

<sup>1</sup>School of Mechanical Engineering, University of Shanghai for Science and Technology, Shanghai, 200093, China.

<sup>2</sup>State Grid Shandong Electric Power Company Dezhou Power Supply Company, Dezhou, 253000, China. ✉e-mail: jilipeng@usst.edu.cn

| Ref. | Method attribution | Method                                                 |                           | The details of Datasets |                 |                                                                                                               |
|------|--------------------|--------------------------------------------------------|---------------------------|-------------------------|-----------------|---------------------------------------------------------------------------------------------------------------|
|      |                    |                                                        |                           | Image size              | Total of images | Defect Types                                                                                                  |
| 7    | Traditional        | HOG + SVM                                              |                           | —                       | 1200            | 4(spot, sand hole, crack, and scratch)                                                                        |
| 8    |                    | Haar-like + extended cascading LogitBoost, DT, and SVM |                           | 4000 × 3000             | 250             | 1(crack)                                                                                                      |
| 17   | Deep Learning      | DCNN                                                   | DCNN + SVM                | 224 × 224               | 171             | 5(crack, paint peeling, scrape, pit, trailing edge damage)                                                    |
| 15   |                    | R-CNN                                                  | Faster R-CNN              | 4000 × 3000             | 458             | 4(LE erosion, VG panel, VG panel with missing teeth, lightning receptor)                                      |
| 18   |                    |                                                        | Image Enhanced Mask R-CNN | 1920 × 1080             | 191             | 4(crack,erosion,void,other)                                                                                   |
| 19   |                    | YOLO                                                   | YSODA (CNN)               | 500 × 375               | 2366            | 3(crack, oil pollution, sand inclusion)                                                                       |
| 16   |                    |                                                        | YOLOv3                    | 5280 × 2970             | 700             | 6(VG panel, VG panel with missing teeth, LE corrosion, crack, lightning detector, damaged lightning detector) |

**Table 1.** Visual detection methods of WT blade defect and related datasets.

were applied to improve recognition performance. Despite these exciting signs of progress, there remains a significant issue of lacking a proper benchmark dataset to meet more challenging problems.

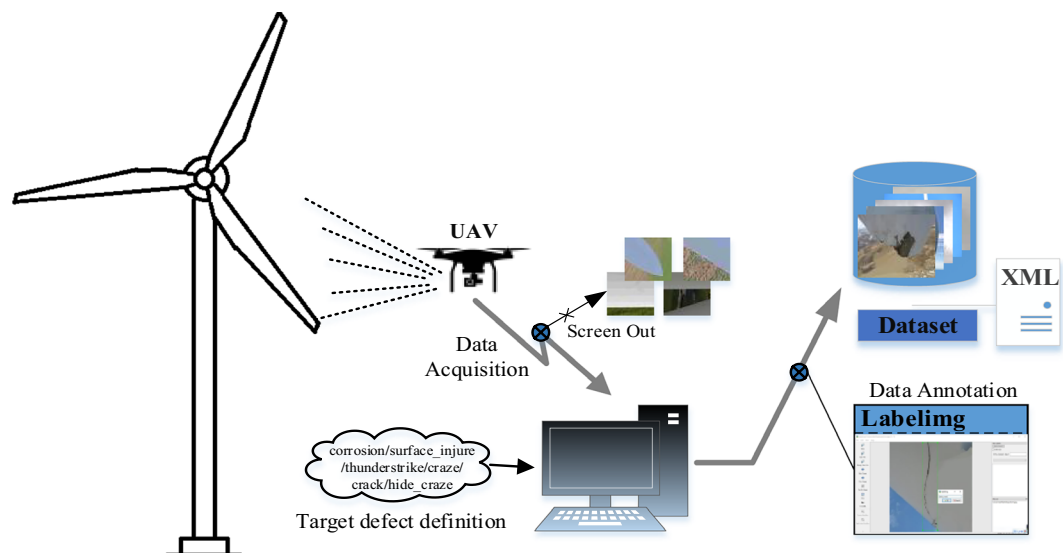
In order to gain a more detailed understanding of the necessity for a new benchmark, a comprehensive comparative study between the proposed dataset and existing datasets is essential. The first major limitation of current datasets is the scarcity of defect types. As highlighted in previous studies<sup>15,16</sup>, there is high diversity in the geometrical and spatial patterns of defects. However, current datasets often fail to capture this complexity<sup>7,8,17–19</sup>. For instance, literature<sup>8</sup> contains only a single defect type (crack), which is insufficient to approximate actual applications compared to the six categories in our proposed dataset. Secondly, regarding dataset size, literature<sup>16,17</sup> contains fewer than 800 images—more than 200 images fewer than our dataset—which hinders data-driven methods that rely on large sample sizes. Finally, regarding learnable visual representations, datasets<sup>15,16</sup> define defect types excessively based on damage location, ignoring the randomness of defects in real scenarios. In contrast, the dataset constructed in this paper considers the distinguishable visual characteristics of each defect type, addressing the urgent demand for a dataset with richer types and more perfect data samples.

With the development of UAV technology, high-definition image acquisition, and labeling technology, favorable conditions are provided for constructing superior WT blade surface defect datasets. The advantages of UAVs in image acquisition<sup>25</sup> have led related studies<sup>7,8,16–19</sup> to use them for dataset construction. Furthermore, the availability of annotation tools like labelImg<sup>26</sup> and Vott<sup>27</sup> allows for more convenient creation of custom datasets and generation of XML files. Against this background, consolidating defect images from different environments is used to optimize dataset representation, thus improving the robustness and generalization of subsequent detection models required for autonomous inspection tasks<sup>28,29</sup>.

This paper proposes the Wind Turbine Blade Defect dataset (WTBD): a new multi-class benchmark WT blade defects dataset. For creating WTBD, a UAV was used to collect images with characteristics such as arbitrary orientations, variations in scale, and irregular shapes. A total of 1,065 images with distinctive defect features were selected, and 1,568 annotated defect instances were divided into 6 categories according to engineering experience and visual feature information. Furthermore, the systematicness and comprehensiveness of WTBD are illustrated through experiments. Specifically, LBP<sup>12</sup> and HOG<sup>13</sup> operators are employed to characterize visual attributes. These traditional descriptors, combined with feature space visualization, validate the dataset's challenging nature, characterized by high intra-class variations and significant inter-class similarity. Consequently, the WTBD serves as a rigorous benchmark resource for developing and evaluating state-of-the-art algorithms in detection, classification, and segmentation tasks.

Our main contributions are as follows:

- (1) Construction of a standardized multiclass wind turbine blade defect dataset (WTBD)  
This paper introduces WTBD, a novel benchmark dataset comprising 1,065 high-resolution UAV-captured images and 1,568 annotated defect instances across six commonly encountered defect types. The dataset captures diverse imaging conditions, including varying scales, angles, and defect co-occurrences, significantly enhancing the data foundation for wind turbine blade defect detection tasks.
- (2) Comprehensive feature space analysis revealing dataset complexity  
The WTBD dataset is systematically analyzed using LBP and HOG feature descriptors combined with t-SNE visualization to demonstrate its high intra-class variability and low inter-class dissimilarity. These properties quantitatively verify that the dataset reflects the complex visual patterns present in real-world inspection scenarios, providing valuable challenges for developing robust detection models.
- (3) Implementation of a rigorous quality control and standardized annotation protocol  
Unlike datasets with ambiguous labels, WTBD employs a strict manual screening process and a double-annotator validation protocol verified by a Cohen's Kappa coefficient of 0.8970. The study establishes a fine-grained taxonomy for six defect types, particularly distinguishing between visually similar categories, ensuring high reliability and consistency for algorithmic benchmarking.



**Fig. 1** The workflow of dataset construction.

## Methods

**Dataset construction procedure.** The construction of the Wind Turbine Blade Defect (WTBD) dataset followed a systematic workflow comprising three primary stages: target defect definition, data acquisition, and data annotation. This structured approach ensures data reliability and traceability. The specific workflow is illustrated in Fig. 1.

**Target defect definition.** Prior to data collection, target defect categories were rigorously defined based on industrial maintenance standards and visual characteristics observable in aerial imagery. Wind turbine blades are subject to varying degrees of external stress, chemical corrosion, and environmental factors during transportation, installation, and operation. These factors result in surface defects with diverse visual manifestations.

By integrating engineering experience with an analysis of low-level visual features (texture, color, edges) and high-level semantic information, we categorized the defects into six distinct classes. Examples of each defect type are presented in Fig. 2.

The specific definitions for the six defect categories are as follows:

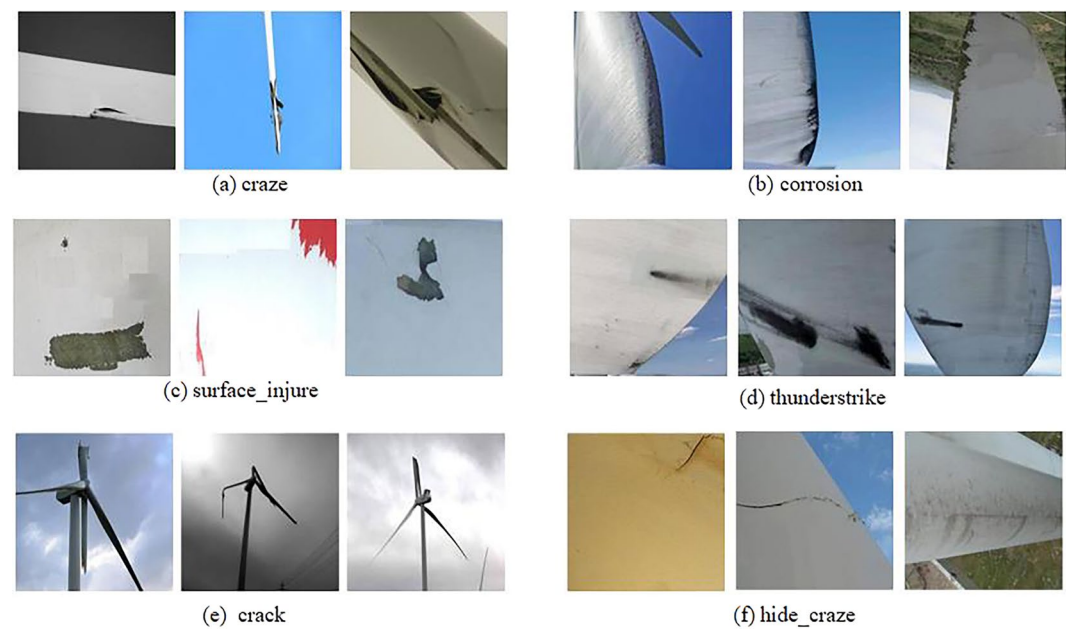
- (1) Craze: Visible cracks and local structural breakage on the blade surface, typically resulting from stress and environmental exposure.
- (2) Corrosion: Dense pitting damage or surface roughness caused by long-term environmental erosion (e.g., sand, salt spray).
- (3) Surface\_injure: Physical damage to the protective coating, such as peeling paint or scratches.
- (4) Thunderstrike: Distinctive damage caused by lightning strikes, characterized by burn marks, superheated discoloration, or structural damage.
- (5) Crack: Deep structural cracking that often extends along the blade body.
- (6) Hide\_craze: Subtle hairline cracks or minor scratches on the outer surface, visually less conspicuous than major crazing.

**Data acquisition and preprocessing.** The raw image data were acquired at a commercial coastal wind farm in Shanghai, China, from May 2023 to August 2023. This location was selected to capture blade defects typical of high-humidity and high-salinity coastal environments. Inspections were conducted using a DJI Phantom 4 Pro V2.0 Unmanned Aerial Vehicle (UAV) equipped with a 1-inch 20-megapixel CMOS sensor.

To capture high-resolution details, the UAV was manually operated at a distance of approximately 6 to 10 meters from the blades under diverse weather conditions (sunny, cloudy, and overcast). Camera settings were strictly controlled (exposure time  $\leq 1/800$  s, ISO 100–400) to ensure image clarity. A comprehensive summary of the acquisition equipment, flight parameters, and environmental conditions is provided in Table 2.

The initial acquisition phase generated approximately 2,500 raw images. We implemented a rigorous manual screening process to filter this raw data based on the following exclusion criteria: (1) significant motion blur or out-of-focus targets; (2) over-exposure or under-exposure rendering defects unidentifiable; and (3) images containing no visible defects (healthy blades).

Following screening, the remaining images were cropped and resized to a standard resolution of  $1024 \times 1024$  pixels to facilitate unified processing, while patches containing no defects were excluded during the screening phase. To strictly maintain data confidentiality and protect the proprietary information of the commercial wind farm, all original metadata linking images to specific turbines or blade serial numbers were anonymized.



**Fig. 2** Examples of samples in WTBD: three pieces with different defects.

| Category          | Parameter           | Description/Value                                   |
|-------------------|---------------------|-----------------------------------------------------|
| Location & Time   | Acquisition Site    | Commercial coastal wind farm, Shanghai, China       |
|                   | Environment Type    | High-humidity and high-salinity coastal environment |
|                   | Dates               | May 2023 – August 2023                              |
| Equipment         | UAV Model           | DJI Phantom 4 Pro V2.0                              |
|                   | Camera Sensor       | 1-inch 20-megapixel CMOS                            |
|                   | Stabilization       | 3-axis gimbal                                       |
| Flight Parameters | Shooting Distance   | 6 – 10 meters (from blade surface)                  |
|                   | Flight Mode         | Manual operation by certified pilots                |
| Imaging Settings  | Exposure Time       | ≤ 1/800 s (to minimize motion blur)                 |
|                   | ISO Sensitivity     | 100 – 400                                           |
|                   | Lighting Conditions | Sunny, cloudy, and overcast                         |
| Data Format       | Raw Image Count     | ~2,500 images                                       |
|                   | Final Resolution    | 1024 × 1024 pixels (standardized)                   |

**Table 2.** Summary of data acquisition parameters and imaging metadata.

Consequently, a simplified sequential numbering scheme (e.g., 0.jpg, 1.jpg, ..., 1064.jpg) was adopted. Although this protocol disassociates image patches from specific turbine coordinates, it is a strict requirement for the public release of this commercial data. This naming convention ensures full compliance with privacy agreements while providing unique, neutral identifiers for algorithmic benchmarking.

**Data annotation and quality control.** To achieve professional-grade ground truth, the dataset was annotated using the open-source graphical image annotation tool labelImg<sup>26</sup>. Annotations were saved in PASCAL VOC (XML) format<sup>30</sup>, recording the bounding box coordinates (x{min}, y{min}, x{max}, y{max}) and the corresponding class label for each defect instance. As shown in Fig. 3, the top-left corner of the image is defined as the coordinate origin. The visualization includes the image size (red box), defect type (blue label), and location information (white bounding box).

To ensure annotation accuracy and consistency, a double-annotator validation process was implemented involving researchers from the School of Mechanical Engineering at the University of Shanghai for Science and Technology and State Grid Shandong Electric Power Company Dezhou Power Supply Company. The second author, who possesses extensive experience in wind turbine blade inspection, performed the primary annotation on all 1,065 images. To assess dataset reliability, the third author, who holds specialized knowledge in defect recognition and power equipment maintenance, independently re-annotated the dataset without viewing the original labels. The consistency between the two annotators was quantified using Cohen's Kappa coefficient. The dataset achieved a Kappa value of 0.8970, indicating a high level of agreement and high-quality defect labels. Any annotation conflicts—such as discrepancies in boundary precision or ambiguity between similar categories

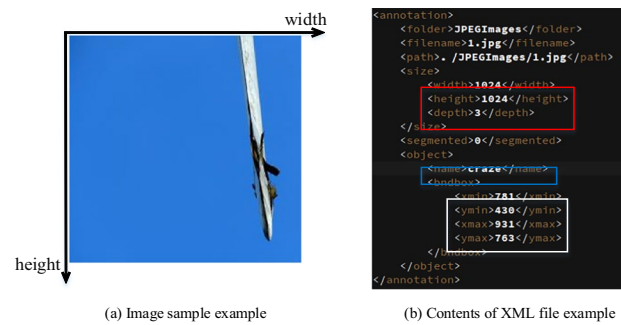


Fig. 3 Annotation example.

| Statistical    | The number of defects | The proportion of the total |
|----------------|-----------------------|-----------------------------|
| craze          | 259                   | 16.50%                      |
| corrosion      | 254                   | 16.20%                      |
| Surface_injure | 394                   | 24.50%                      |
| thunderstrike  | 92                    | 5.90%                       |
| crack          | 224                   | 14.30%                      |
| hide_craze     | 345                   | 22.00%                      |

Table 3. The number of different defect types in WTBD.

like “craze” and “hide\_craze”—were resolved through a consensus meeting with the corresponding author to determine the final ground truth.

After the rigorous annotation and quality control process (verified by a Cohen’s Kappa of 0.8970), the final dataset composition was established, as detailed in Table 3. The WTBD dataset consists of 1,065 blade images; however, the total number of annotated defect instances is 1,568. This discrepancy arises because multiple defect types often co-occur within a single image, reflecting the complexity of real-world damage accumulation. As shown in Table 3, the class distribution exhibits a natural imbalance consistent with field inspection experience: ‘surface\_injure’ is the most frequent category (24.5%), while severe structural damages like ‘thunderstrike’ are rare (5.9%). This imbalance preserves the realistic characteristics of wind farm environments, posing a meaningful challenge for the robustness of subsequent detection models.

**Ethical approval.** This article does not contain any studies with human participants or animals performed by any of the authors.

Data Records

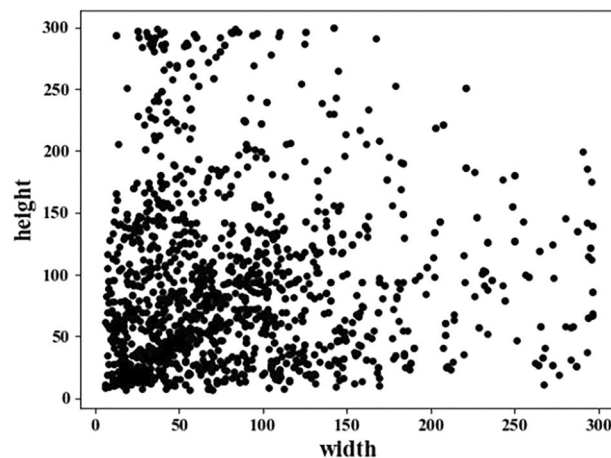
The Wind Turbine Blade Defect (WTBD) dataset is publicly available at Figshare<sup>31</sup> under the identifier DOI: 10.6084/m9.figshare.30210175. The repository is systematically organized into two primary directories: Images and Annotations. The Images directory contains 1,065 high-resolution JPEG images, which have been standardized to a resolution of 1024 × 1024 pixels. As detailed in the Methods section, filenames within this directory utilize a simplified sequential numerical index (e.g., 0.jpg, 1.jpg, ..., 1064.jpg) to strictly maintain data anonymity while providing unique identifiers for benchmarking. Correspondingly, the Annotations directory comprises 1,065 annotation files stored in PASCAL VOC XML format. Each XML file shares the identical filename prefix as its associated image (e.g., 0.xml) and encapsulates precise bounding box coordinates (x{min}, y{min}, x{max}, y{max}) alongside the specific defect category labels (e.g., crack, thunderstrike) for every annotated instance. All data files are open access and can be downloaded directly for research purposes. Researchers are encouraged to split the dataset into training, validation, and testing sets (e.g., a typical 70%/15%/15% ratio) ensuring that data splits are randomized to prevent bias.

Technical Validation

To further explore the information on the defect areas in WTBD, the target frame sizes within the blade defect dataset are counted. All sample defect frames were normalized, and the width and height values were uniformly adjusted to 300 × 300. The statistical performance of the target frame size is shown in Fig. 4. It can be seen that the number of defects decreases as the defect target frame area increases, indicating that more minor target blade defects are more frequent and common under normal inspection operations at wind farms.

Beyond the statistical distribution of target sizes, we further validated the dataset’s technical quality by assessing its visual complexity and alignment with real-world inspection scenarios. This visual validation focuses on three critical aspects:





**Fig. 4** Multi-scale defect target box statistics.

- (1) Higher intra-class variations: In the aerial image of the UAV, due to the high-resolution camera, the structure and details of the WT blade are more apparent to a certain extent, which brings certain conditions for the detection of defective targets. Firstly, because of the freedom of UAV image acquisition, the visual representation of the same defect differs at different heights, distances, and angles. Secondly, WTs are located in various outdoor environments. The camera is easily affected by complex environmental factors such as light, which will cause differences in the final collection results of the same defect sample and increase the intra-class diversities of WTBD. As shown in Fig. 5(a), the visual performance differs from different backgrounds and shooting angles for the three pictures under the craze defect. Therefore, it can be considered that WTBD has high intra-class diversity of defects, which can enhance the robustness of the WT blade surface defect detection model to a certain extent.
- (2) Smaller inter-class dissimilarity: Under the actual task of WT blade defect detection, there may be little visual distinction between the different types of defects. Also, considering that the difference between the background elements in the WT blade images of varying defect types may be slight, the WT blade images containing the targets of different defect categories under similar backgrounds are added when constructing the WTBD. In this way, the state of WTBD is as close as possible to the real WT blade defect detection task, thereby enhancing the detection ability of the model. As shown in Fig. 5(b), in the example images of defect types corrosion and thunderstrike, there are visual features with blue sky as the background and a relatively large area of blade elements. Furthermore, the corrosion and thunderstrike images show some degree of blade abrasion in the defective regions.
- (3) Multi-class defects and multi-scale targets: Compared to previous studies, the defect types in WTBD are more comprehensive, and the division into six defect types makes defect detection more refined and systematic. At the same time, there are multiple defect areas of the same or different types on a single wind turbine blade image to ensure as much diversity as possible in the collected data. As shown in Fig. 5(c), there may be multiple thunderstrike samples and surface\_injure and hide\_craze co-existing samples in WTBD, which is consistent with the experience of manual inspection in an engineering environment. Besides, the defect marking frames exhibit multi-scale phenomena due to the different causes and periods of each type of defect, as well as the viewpoint and distance of the UAV during filming. As shown in Fig. 4, at a normalized scale of  $300 \times 300$ , the dimensions of the defective target boxes exhibit dispersion.

To assess the quality, complexity, and distinctiveness of the WTBD dataset independent of specific deep learning architectures, we conducted a feature space analysis using traditional computer vision descriptors. This approach validates the dataset's diversity and highlights the challenges effective detection models must overcome, such as distinguishing visually similar defects and handling varying imaging conditions.

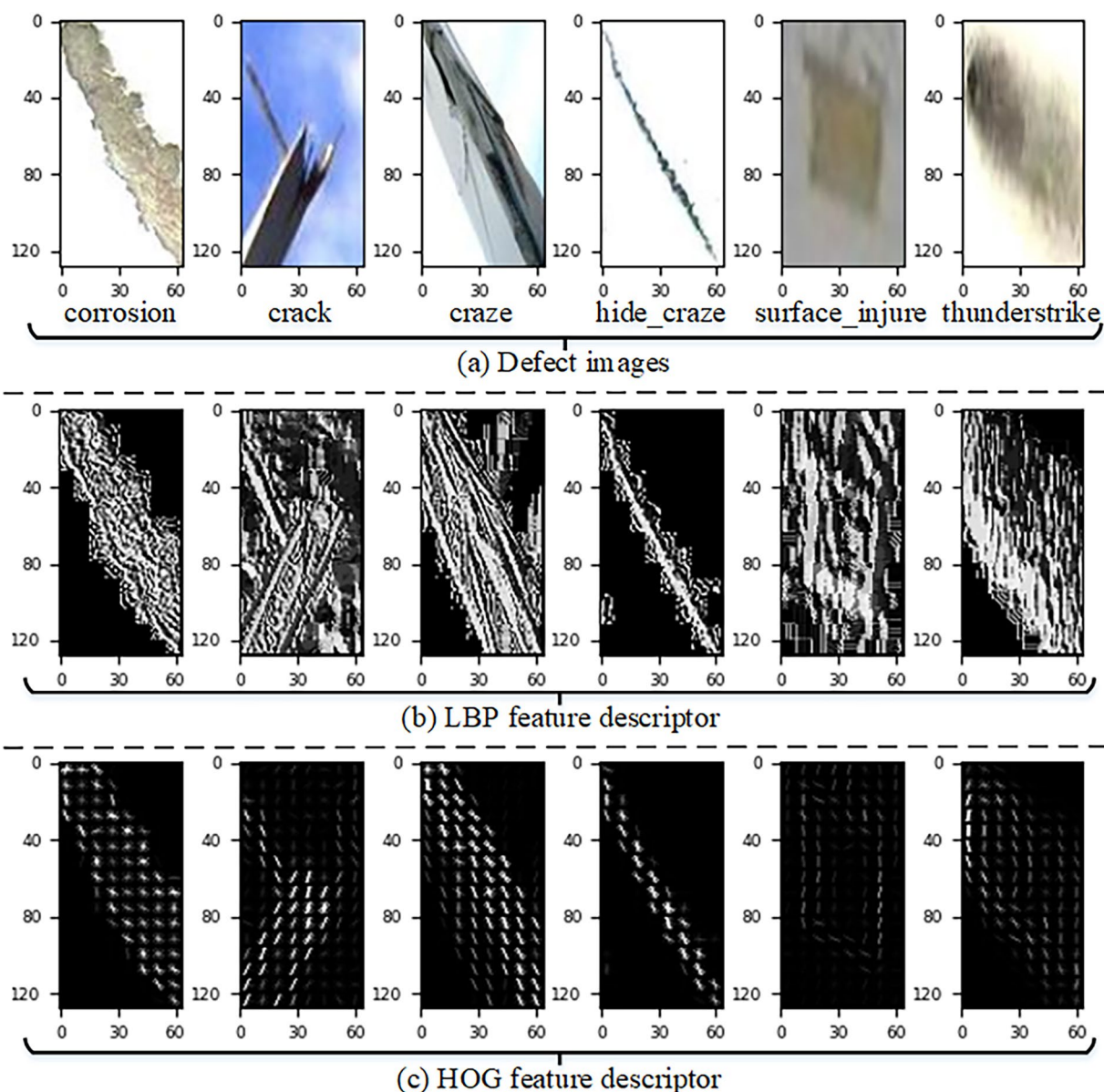
**Feature Extraction and Visual Analysis** For each annotated defect instance, the region of interest (ROI) was cropped based on the ground truth bounding boxes. We extracted two types of features to represent the visual properties of the defects: Local Binary Patterns (LBP) to characterize texture irregularities, and Histogram of Oriented Gradients (HOG) to capture local shape and edge directionality.

As illustrated in Fig. 6, the transformed feature maps reveal varying degrees of local pattern and edge complexities. While these descriptors capture basic defect attributes, the visual representations also highlight potential detection challenges. For instance, in the cases of surface\_injure and thunderstrike, the feature maps exhibit ambiguous information within the defect contour closure regions. This visual ambiguity suggests that relying solely on low-level features may lead to classification confusion, indicating a need for more advanced feature learning.

Furthermore, the dataset exhibits significant variations even within the same defect category. Figure 7 displays three different samples of the craze defect. It is evident that despite belonging to the same class, these samples possess distinct external contour structures and texture orientations in their feature maps. This implies a high degree of intra-class variability, driven by the random nature of defect formation and varying imaging angles.



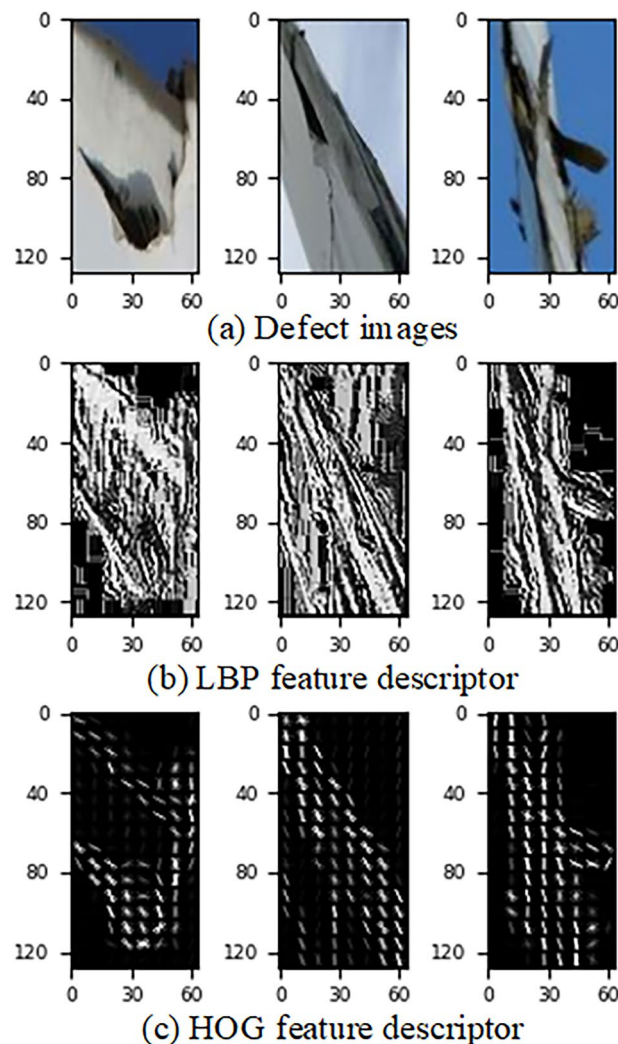
**Fig. 5** Examples of descriptions for dataset properties.



**Fig. 6** Example of feature maps for each type of defective sample under the LBP and HOG feature description operators.

To quantify the complexity of the dataset and demonstrate the necessity of deep feature learning, we employed t-Distributed Stochastic Neighbor Embedding (t-SNE) to project the high-dimensional HOG and LBP feature vectors into a two-dimensional manifold. This visualization provides a qualitative assessment of the difficulty of the classification boundaries.

The t-SNE projections presented in Fig. 8 reveal a highly entangled feature space, substantiating the high challenge level of the WTBD dataset:



**Fig. 7** Schematic diagram of craze defect conversion with two feature operators.

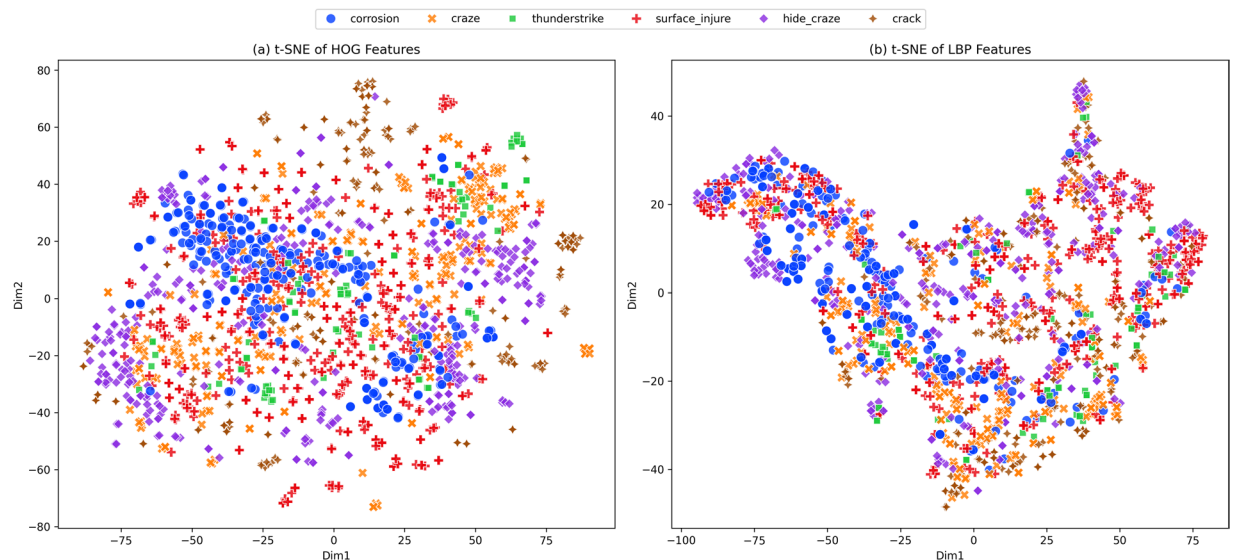
- (1) High Intra-class Variability (Dispersion): As observed in Fig. 8(b), samples belonging to the same category do not form compact, isolated clusters but are instead widely distributed across the manifold. For instance, the craze class (orange crosses) spans a broad region of the feature space. This dispersion indicates that defects of the same type exhibit significant visual heterogeneity—varying drastically in scale, aspect ratio, and texture depending on the viewing angle and lighting conditions. This validates that the dataset captures the complex, non-standardized nature of real-world defects.
- (2) Severe Inter-class Similarity (Entanglement): Fig. 8(a) demonstrates a significant semantic overlap between different defect types. Specifically, the features for corrosion (blue circles) and surface\_injure (red pluses) are heavily intertwined, forming a dense, inseparable core. This indicates that these defects share highly similar low-level gradients and edge properties, making them indistinguishable using shallow feature descriptors. Similarly, the thunderstrike samples (green squares) are scattered among other categories, suggesting that their visual signature often mimics other surface damages.

The lack of distinct decision boundaries in the traditional feature space (HOG/LBP) confirms that the WTBD dataset is non-trivial. It proves that the visual distinction between these defect types relies on high-level semantic context rather than simple texture or edge primitives. Consequently, this dataset serves as a rigorous benchmark for evaluating deep learning models, which must possess strong generalization capabilities to resolve these complex visual entanglements.

### Usage Notes

To comply with a strict Non-Disclosure Agreement (NDA) with the commercial wind farm operator, all metadata linking images to specific turbine serial numbers or geographic coordinates has been removed. Consequently, filenames utilize a randomized sequential index (e.g., 0.jpg, 1.jpg, ..., 1064.jpg). Beyond privacy compliance, this randomized naming protocol is intentional from a machine learning perspective: it





**Fig. 8** t-SNE visualization of defect feature distributions in the WTBD dataset. **(a)** Distribution based on HOG features (shape/edge). **(b)** Distribution based on LBP features (texture). The dense overlap and lack of linear separability between categories highlight the complexity of the dataset.

disassociates defect samples from specific turbine contexts, encouraging models to learn robust, generalized defect features rather than overfitting to the background or lighting conditions of a single turbine.

To facilitate standardized benchmarking and reproducibility, we strictly recommend splitting the dataset into training (70%), validation (15%), and testing (15%) sets. Users are encouraged to utilize the standard split index file (train\_val\_test\_split.txt) provided in the accompanying code repository to ensure valid comparisons with future studies.

Users should be aware that the dataset reflects real-world inspection frequencies. Certain categories (e.g., thunderstrike) contain relatively few samples compared to common defects like surface\_injure. Researchers may consider data augmentation or resampling techniques to address this class imbalance during training. Additionally, the dataset contains multi-scale and multi-class defect instances co-occurring within single images, providing a rigorous test bed for developing models capable of handling complex field scenarios. While images containing no defects were excluded, the retained annotated images contain significant non-defect background areas. These regions naturally serve as negative samples during object detection training, aiding in the reduction of false positives<sup>32–35</sup>.

### Data availability

The dataset has been deposited in Figshare. The custom code used for data quality validation (calculate\_kappa.py), feature extraction (LBP/HOG) and t-SNE visualization (tsne\_analysis.py) is available within the same Figshare repository (<https://doi.org/10.6084/m9.figshare.30210175>)<sup>31</sup>. A demonstration script (preprocessing\_demo.py) replicating the standard image resizing and normalization procedure is also included. The standard data split index file (train\_val\_test\_split.txt) is provided to ensure reproducibility. Standard open-source libraries (e.g., PyTorch, scikit-learn) were used for benchmarking as detailed in the Methods section.

Received: 7 October 2025; Accepted: 29 January 2026;

Published online: 09 February 2026

### References

- McKenna, R. *et al.* System impacts of wind energy developments: Key research challenges and opportunities. *Joule* **9**, 1 (2025).
- Zhang, S. *et al.* UAV based defect detection and fault diagnosis for static and rotating wind turbine blade: a review. *Nondestruct. Test. Eval.* **40**, 1691–1729 (2025).
- Gohar, I., Yew, W. K., Halimi, A. & See, J. Review of state-of-the-art surface defect detection on wind turbine blades through aerial imagery: Challenges and recommendations. *Eng. Appl. Artif. Intell.* **144**, 109970 (2025).
- Dodge, S. & Karam, L. Understanding how image quality affects deep neural networks. In *2016 Eighth International Conference on Quality of Multimedia Experience (QoMEX)* 1–6 (IEEE, 2016).
- Zhu, X. *et al.* Damage identification of wind turbine blades based on deep learning and ultrasonic testing. *Nondestruct. Test. Eval.* **40**, 508–533 (2025).
- Kim, J. & Lee, S. Deep learning of human visual sensitivity in image quality assessment framework. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 1676–1684 (2017).
- Deng, L., Guo, Y. & Chai, B. Defect detection on a wind turbine blade based on digital image processing. *Processes* **9**, 1452 (2021).
- Wang, L. & Zhang, Z. Automatic detection of wind turbine blade surface cracks based on UAV-taken images. *IEEE Trans. Ind. Electron.* **64**, 7293–7303 (2017).
- Du, Y. *et al.* Damage detection techniques for wind turbine blades: A review. *Mech. Syst. Signal Process.* **141**, 106445 (2020).
- Chen, Y. *et al.* Surface defect detection methods for industrial products: A review. *Appl. Sci.* **11**, 7657 (2021).
- Tang, B., Chen, L., Sun, W. & Lin, Z. K. Review of surface defect detection of steel products based on machine vision. *IET Image Process.* **17**, 303–322 (2023).

12. Ojala, T., Pietikainen, M. & Maenpää, T. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Trans. Pattern Anal. Mach. Intell.* **24**, 971–987 (2002).
13. Dalal, N. & Triggs, B. Histograms of oriented gradients for human detection. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)* Vol. 1, 886–893 (2005).
14. Platt, J. C. Sequential minimal optimization: A fast algorithm for training support vector machines. (Microsoft, 1998).
15. Shihavuddin, A. S. M. *et al.* Wind turbine surface damage detection by deep learning aided drone inspection analysis. *Energies* **12**, 676 (2019).
16. Sarkar, D. & Gunturi, S. K. Wind turbine blade structural state evaluation by hybrid object detector relying on deep learning models. *J. Ambient Intell. Humaniz. Comput.* **12**, 8535–8548 (2021).
17. Yu, Y., Cao, H., Yan, X., Wang, T. & Ge, S. S. Defect identification of wind turbine blades based on defect semantic features with transfer feature extractor. *Neurocomputing* **376**, 1–9 (2020).
18. Zhang, J., Cosma, G. & Watkins, J. Image enhanced mask R-CNN: A deep learning pipeline with new evaluation measures for wind turbine blade defect detection and classification. *J. Imaging* **7**, 46 (2021).
19. Qiu, Z., Wang, S., Zeng, Z. & Yu, D. Automatic visual defects inspection of wind turbine blades via YOLO-based small object detection approach. *J. Electron. Imaging* **28**, 043023 (2019).
20. Ren, S., He, K., Girshick, R. & Sun, J. Faster R-CNN: Towards real-time object detection with region proposal networks. *Adv. Neural Inf. Process. Syst.* **28** (2015).
21. He, K., Gkioxari, G., Dollár, P. & Girshick, R. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision* 2961–2969 (2017).
22. Liu, W. *et al.* SSD: Single shot multibox detector. In *Computer Vision—ECCV 2016* 21–37 (Springer, 2016).
23. Redmon, J. & Farhadi, A. YOLOv3: An incremental improvement. *arXiv preprint arXiv:1804.02767* (2018).
24. Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision* 2980–2988 (2017).
25. Miao, J. G., Wang, J. Y., Zhang, H. & Miao, Q. Review of the development of fault diagnosis technology for unmanned aerial vehicle. *Chin. J. Sci. Instrum.* **41**, 56–69 (2020).
26. Tzutalin, D. LabelImg. *GitHub* <https://github.com/tzutalin/labelImg> (2015).
27. Microsoft. VoTT (Visual Object Tagging Tool). *GitHub* <https://github.com/microsoft/VoTT>.
28. Ma, M. *et al.* Fully autonomous inspection of wind turbine blades based on drones and artificial intelligence. *Electr. Age* **S2**, 44–48 (2023).
29. Peng, L. & Liu, J. Detection and analysis of large-scale WT blade surface cracks based on UAV-taken images. *IET Image Process.* **12**, 2059–2064 (2018).
30. Everingham, M. *et al.* The Pascal Visual Object Classes Challenge: A retrospective. *Int. J. Comput. Vis.* **111**, 98–136 (2015).
31. Ji, L., Cheng, J., & Wu, S. Multiclass Dataset for Intelligent Detection of Wind Turbine Blade Defects Using Drone Imagery, *figshare*, <https://doi.org/10.6084/m9.figshare.30210175> (2026).
32. Simonyan, K. & Zisserman, A. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations* (2015).
33. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 770–778 (2016).
34. Lin, T.-Y. *et al.* Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* 2117–2125 (2017).
35. Lin, T.-Y. *et al.* Microsoft COCO: Common objects in context. In *Computer Vision—ECCV 2014* 740–755 (Springer, 2014).

## Acknowledgements

This work was supported in part by “Science and Technology Innovation Action Plan” Artificial Intelligence Science and Technology Support Special Project of Shanghai (20511101600) and Shanghai Professional Technical Service Platform for Intelligent Operation and Maintenance of Renewable Energy (22DZ2291800). This work was supported in part by NSFC (62003212).

## Author contributions

L.J. (first and corresponding author) conceived and designed the study, supervised the project, and contributed to manuscript revision. J.C. (second author) contributed to dataset collection, annotation, and technical validation experiments. S.W. (third author) contributed to data analysis and annotation. All authors discussed the results, reviewed, and approved the final version of the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

**Correspondence** and requests for materials should be addressed to L.J.

**Reprints and permissions information** is available at [www.nature.com/reprints](http://www.nature.com/reprints).

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026